10 Useful Sudoers Configurations for Setting 'sudo' in Linux

Aaron Kili | Last Updated: July 13, 2023 | Linux Commands | 11 Comments

In Linux and other Unix-like operating systems, only the **root** user can run all commands and perform certain critical operations on the system such as install and update, remove packages, create users and groups, modify important system configuration files and so on.

However, a system administrator who assumes the role of the root user can permit other normal system users with the help of sudo command and a few configurations to run some commands as well as carry out a number of vital system operations including the ones mentioned above.

Alternatively, the system administrator can share the root user password (which is not a recommended method) so that normal system users have access to the root user account via **su** command.

sudo allows a permitted user to execute a command as root (or another user), as specified by the security policy:

1. It reads and parses **/etc/sudoers**, looks up the invoking user and its permissions,
2. then prompts the invoking user for a password (normally the user's password, but it can as well be the target user's password. Or it can be skipped with NOPASSWD tag),
3. after that, sudo creates a child process in which it calls **setuid()** to switch to the target user
4. next, it executes a shell or the command given as arguments in the child process above.

Below are ten **/etc/sudoers** file configurations to modify the behavior of **sudo** command using **Defaults** entries.

```
$ sudo cat /etc/sudoers
```

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

/etc/sudoers File

```
#
# This file MUST be edited with the 'visudo' command as root
#
# Please consider adding local content in /etc/sudoers.d/ ins
# directly modifying this file.
#
# See the man page for details on how to write a sudoers file
#
Defaults        env_reset
Defaults        mail_badpass
Defaults        secure_path="/usr/local/sbin:/usr/local/bin:/
Defaults        logfile="/var/log/sudo.log"
Defaults        lecture="always"
Defaults        badpass_message="Password is wrong, please tr
Defaults        passwd_tries=5
Defaults        insults
Defaults        log_input,log_output
```

Types of Defaults Entries

```
Defaults                parameter,   parameter_list     #affe
Defaults@Host_List      parameter,   parameter_list     #affe
Defaults:User_List      parameter,   parameter_list     #affe
Defaults!Cmnd_List      parameter,   parameter_list     #affe
Defaults>Runas_List     parameter,   parameter_list     #affe
```

For the scope of this guide, we will zero down to the first type of **Defaults** in the forms below. Parameters may be flags, integer values, strings, or lists.

You should note that flags are implicitly boolean and can be turned off using the `'!'` operator, and lists have two additional assignment operators, `+=` (add to list) and `-=` (remove from list).

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

```
Defaults     parameter
OR
Defaults     parameter=value
OR
Defaults     parameter -=value
Defaults     parameter +=value
OR
Defaults     !parameter
```

1. Set a Secure PATH

This is the path used for every command run with sudo, it has two importances:

1. Used when a system administrator does not trust sudo users to have a secure PATH environment variable
2. To separate “root path” and “user path”, only users defined by **exempt_group** are not affected by this setting.

To set it, add the line:

```
Defaults secure_path="/usr/local/sbin:/usr/local/bin:/usr/sb
```

2. Enable sudo on TTY User Login Session

To enable sudo to be invoked from a real **tty** but not through methods such as **cron** or **cgi-bin** scripts, add the line:

```
Defaults  requiretty
```

3. Run Sudo Command Using a pty

A few times, attackers can run a malicious program (such as a virus or malware) using sudo, which would again fork a background process that remains on the user’s

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

terminal device even when the main program has finished executing.

To avoid such a scenario, you can configure sudo to run other commands only from a **psuedo-pty** using the `use_pty` parameter, whether I/O logging is turned on or not as follows:

```
Defaults  use_pty
```

4. Create a Sudo Log File

By default, sudo logs through syslog(3). However, to specify a custom log file, use the logfile parameter like so:

```
Defaults  logfile="/var/log/sudo.log"
```

To log hostname and the four-digit year in the custom log file, use **log_host** and **log_year** parameters respectively as follows:

```
Defaults  log_host, log_year, logfile="/var/log/sudo.log"
```

Below is an example of a custom sudo log file:

```
aaronkilik@tecmint ~ $ cat /var/log/sudo.log
Dec 28 11:55:04 : aaronkilik : TTY=pts/0 ; PWD=/home/aaronkilik ; USER=root ;
    COMMAND=/usr/sbin/visudo
Dec 28 12:03:19 : aaronkilik : TTY=pts/0 ; PWD=/home/aaronkilik ; USER=root ;
    COMMAND=/usr/sbin/visudo
Dec 28 12:03:44 : aaronkilik : TTY=pts/0 ; PWD=/home/aaronkilik ; USER=root ;
    COMMAND=/usr/sbin/visudo
Dec 28 12:04:01 : aaronkilik : TTY=pts/0 ; PWD=/home/aaronkilik ; USER=root ;
    COMMAND=/usr/sbin/visudo
Dec 28 12:04:39 : aaronkilik : TTY=pts/1 ; PWD=/home/aaronkilik ; USER=root ;
    COMMAND=/usr/sbin/visudo
Dec 28 12:05:58 : aaronkilik : 2 incorrect password attempts ; TTY=pts/0 ;
    PWD=/home/aaronkilik ; USER=root ; COMMAND=/usr/sbin/visudo
Dec 28 12:06:58 : aaronkilik : 3 incorrect password attempts ; TTY=pts/0 ;
    PWD=/home/aaronkilik ; USER=root ; COMMAND=/usr/sbin/visudo
Dec 28 12:21:02 : aaronkilik : 1 incorrect password attempt ; TTY=pts/0 ;
    PWD=/home/aaronkilik ; USER=root ; COMMAND=/usr/sbin/visudo
Dec 28 12:46:54 : aaronkilik : TTY=pts/1 ; PWD=/home/aaronkilik ; USER=root ;
```

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

Create Custom Sudo Log File

5. Log Sudo Command Input/Output

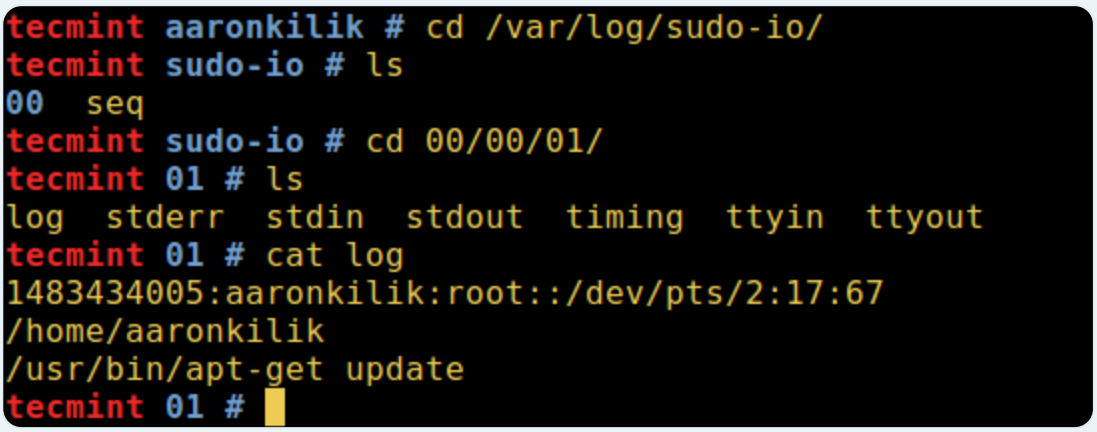
The **log_input** and **log_output** parameters enable sudo to run a command in pseudo-tty and log all user input and all output sent to the screen receptively.

The default I/O log directory is **/var/log/sudo-io**, and if there is a session sequence number, it is stored in this directory. You can specify a custom directory through the **iolog_dir** parameter.

```
Defaults   log_input, log_output
```

There are some escape sequences are supported such as `%{seq}` which expands to a monotonically increasing base-36 sequence number, such as 000001, where every two digits are used to form a new directory, e.g. **00/00/01** as in the example below:

```
$ cd /var/log/sudo-io/
$ ls
$ cd  00/00/01
$ ls
$ cat log
```



Log sudo Input Output

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

You can view the rest of the files in that directory using the cat command.

6. Lecture Sudo Users

To lecture sudo users about password usage on the system, use the **lecture** parameter as below.

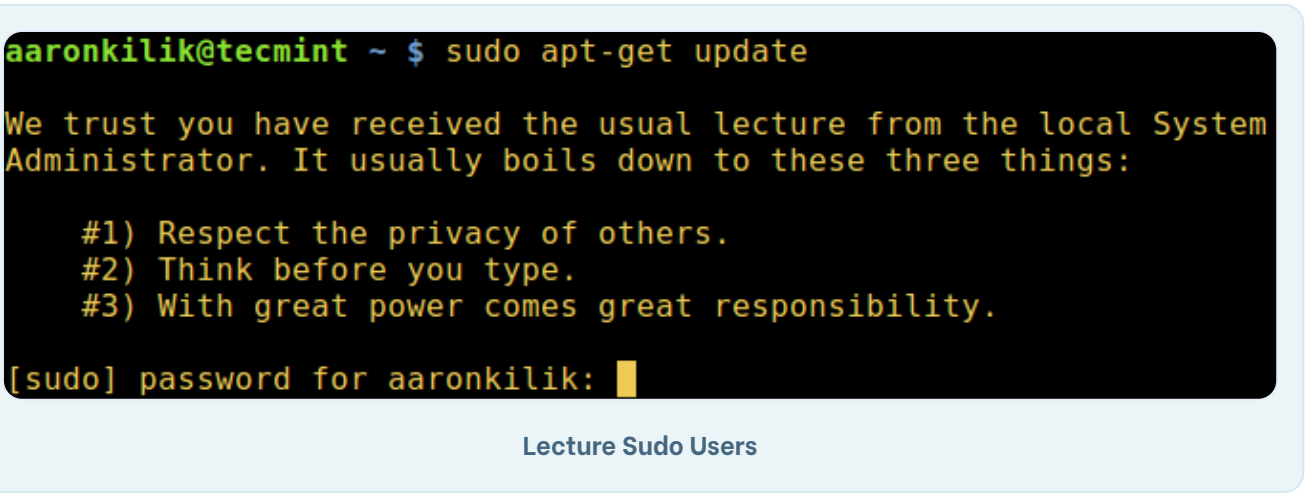
It has 3 possible values:

1. always – always lecture a user.
2. once – only lecture a user the first time they execute sudo command (this is used when no value is specified)
3. never – never lecture the user.

```
Defaults  lecture="always"
```

Additionally, you can set a custom lecture file with the **lecture_file** parameter, type the appropriate message in the file:

```
Defaults  lecture_file="/path/to/file"
```



Lecture Sudo Users

7. Show Custom Message When You Enter Wrong sudo Password

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

When a user enters a wrong password, a certain message is displayed on the command line. The default message is "**sorry, try again**", you can modify the message using the **badpass_message** parameter as follows:

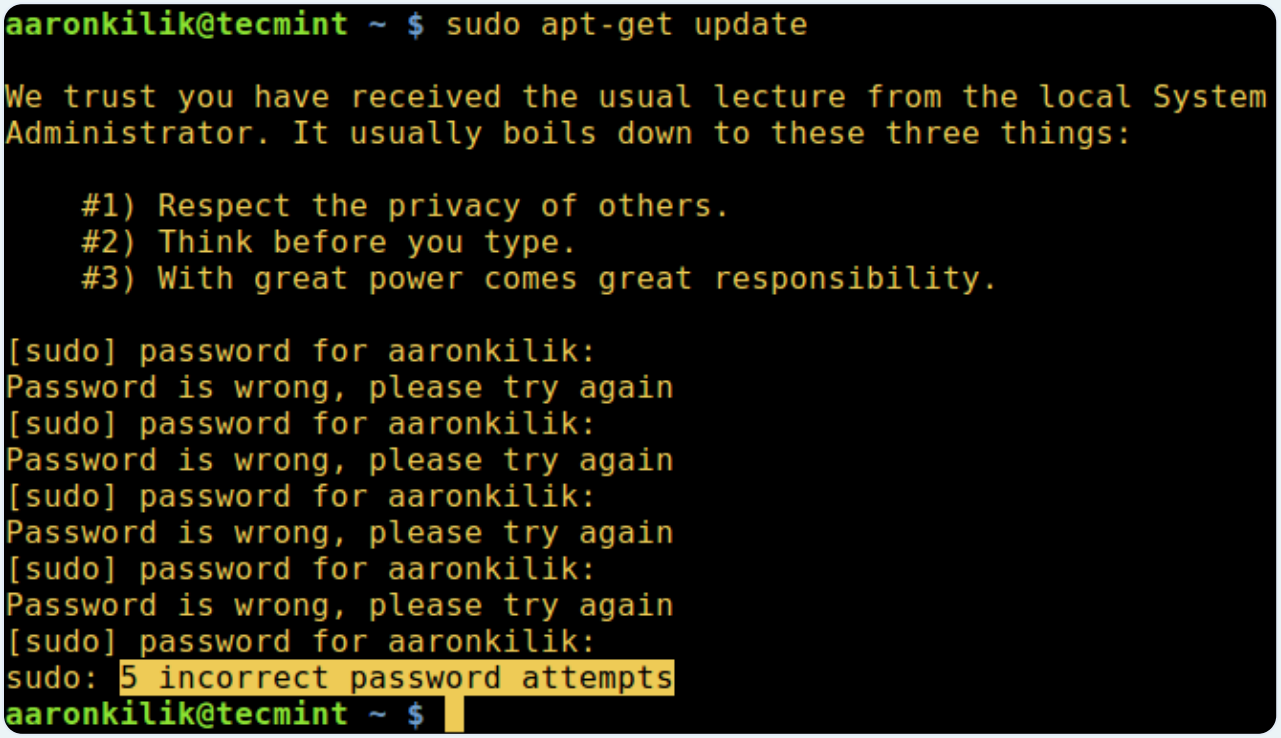
```
Defaults  badpass_message="Password is wrong, please try aga
```

8. Increase sudo Password Tries Limit

The parameter **passwd_tries** is used to specify the number of times a user can try to enter a password.

The default value is 3:

```
Defaults   passwd_tries=5
```



Increase Sudo Password Attempts

To set a password timeout (default is 5 minutes) using **passwd_timeout** parameter, add the line below:

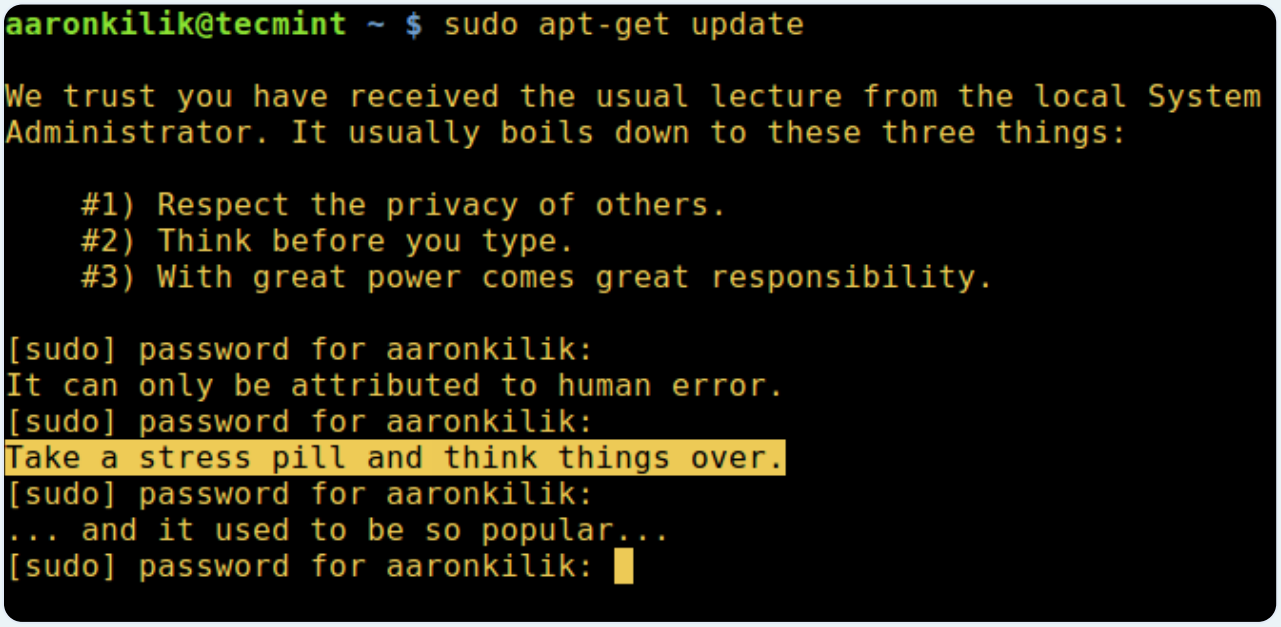
Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

```
Defaults    passwd_timeout=2
```

9. Let Sudo Insult You When You Enter Wrong Password

In case a user types a wrong password, sudo will display insults on the terminal with the insults parameter. This will automatically turn off the **badpass_message** parameter.

```
Defaults  insults
```



Let's Sudo Insult You When Enter Wrong Password

Read More: Let Sudo Insult You When You Enter Incorrect Password

10. Learn More Sudo Configurations

Additionally, you can learn more **sudo** command configurations by reading: Difference Between su and sudo and How to Configure sudo in Linux.

That's it! You can share other useful sudo command configurations or tricks and tips with Linux users out there via the comment section below.

Linux Tricks, sudo commands

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

PREVIOUS ARTICLE:

[12 Useful Commands For Filtering Text for Effective File Operations in Linux]

NEXT ARTICLE:

[3 Ways to Permanently and Securely Delete 'Files and Directories' in Linux]



Aaron Kili

Aaron Kili is a Linux and F.O.S.S enthusiast, an upcoming Linux SysAdmin, web developer, and currently a content creator for TecMint who loves working with computers and strongly believes in sharing knowledge.

*Each tutorial at **TecMint** is created by a team of experienced Linux system administrators* so that it meets our high-quality *standards.*

Join the TecMint Weekly Newsletter (More Than **156,129** Linux Enthusiasts Have Subscribed)

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

Was this article helpful? Please add a comment or buy me a coffee to show your appreciation.

11 thoughts on "10 Useful Sudoers Configurations for Setting 'sudo' in Linux"

Ashok
September 15, 2021 at 12:26 am

Hello Aaron, thank you for sharing this info on sudo configuration. Is there a way for sudo to record **tty** commands run by a ldap user after sudo to a service account and store them to a file?

Example:

```
[user@test1] / # sudo su - oracle
[oracle@test1] /home/oracle # hostname
test1
[oracle@test1] /home/oracle # pwd
/home/oracle
```

How would I let sudo store these commands (hostname, pwd, etc) to a file after a user sudo to service account (oracle, etc) and run commands as that service account on a server? I have an audit requirement on AIX 7.2 servers to capture all TTY session commands run by a user after sudo to a service account hosted on each server.

Thank you for your time.

Thank you.

Reply

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

Admin



Ravi Saive

September 15, 2021 at 10:40 am

@Ashok,

First, create a log directory and set the sticky bit on it.

```
# mkdir -p /var/log/users_historylogs/
# chmod +t  /var/log/users_historylogs/
```

Next, create a new script file under **/etc/profile.d/** directory.

```
# vi /etc/profile.d/history_log.sh
```

And add the below content at the bottom, save, and exit.

```
_who_am_i=$(whoami|awk '{print $1}')
_ID=$(id -u $_who_am_i)

if [ "$_ID" > 0 ]
then
export HISTSIZE=10000
export HISTTIMEFORMAT='%F %T '
export HISTFILE=/var/log/users_historylogs/history-
users-$(whoami | awk '{print $1}';exit)-$(date +%F)
export PROMPT_COMMAND='history -a'
fi
```

Set the permission and enable the script.

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

```
# chmod 770 /etc/profile.d/history_log.sh
# source /etc/profile.d/history_log.sh
```

Now all user executed commands history saved to log file...

Reply

Ashok B
September 16, 2021 at 4:18 am

Hi Ravi,

Thank you so much for the quick response. Does the same procedure apply on AIX 7.2 servers since it uses Korn shell by default? I see that the **/etc/profile.d** directory does not exist on AIX servers and I have implemented step 1 by adding those lines in **/etc/profile** to set up secondary logging for root which looks good as the file was created, not sure if I should add commands to the same **/etc/profile** file to setup secondary logging for other users except root. Thank you.

Reply

Admin



Ravi Saive
September 16, 2021 at 11:31 am

@Ashok,

I think you can add those lines to /etc/profile and see if it is working or not...

Reply

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

Alajjana

February 10, 2023 at 1:37 am

This is a minor typo.

```
$(who am i | awk '{print $1}';exit)
```

should be:

```
$(whoami | awk '{print $1}';exit)
```

Fantastic. Thank you for this script.

Reply

Admin



Ravi Saive

February 10, 2023 at 12:28 pm

@Aljjana,

Thanks, corrected the command in the script...

Reply

Ryan Quezada G.

March 22, 2020 at 9:42 pm

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

This is an excellent article. With this, I complement my knowledge about the root users and permissions.

Reply



Aaron Kili

March 25, 2020 at 1:51 pm

@Ryan

Great! We are grateful that this has helped you gain more knowledge about Linux. Many thanks for the useful feedback.

Reply

Brain

November 11, 2018 at 8:05 pm

Hi,

Thanks for the nice overview.

Can you help with this?

I want to mount a special source without root privileges. So I made an entry in the **/ect/sudoers** file:

```
username        ALL = NOPASSWD: /sbin/mount.cifs,
/bin/umount /mnt/folder
```

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

How can I restrict the source that I want to mount to be only one that can be mounted. Now username can mount everything.

Thanks in advance.

Bye,

Reply

Garry Garrett
August 14, 2019 at 7:45 pm

I think what you may want to do is, instead of using **sudo**, add the mount to **/etc/fstab**, and include the option **"user"** (see the man page on "mount"). What this will do is allow ordinary users to mount/unmount the filesystem. That would allow ALL users to mount/unmount it. They can then say "**mount /mnt/folder**". This mount option is specific to Linux and would not work on other flavors of Unix.

Another option would be to use the automounter. You could setup a direct automount map. Then whenever a user does "**cd /mnt/folder**", it mounts. After it mounts, every 5 minutes, it half-heartedly attempts to unmount it, which will not be successful if it is still in use. Again, this would allow ALL users to mount it.

If you really want just the one user to be able to mount/unmount, then you'd need to spell out the full mount command (not **/sbin/mount.cifs**):

```
user ALL = NOPASSWD: /bin/mount /path-to-device /mnt/folder,
/bin/umount /mnt/folder
```

(there might be some options you'll want to specify after "mount", e.g. "-o ro", "-t cifs", etc.). The user will then need to type the command-line

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com

exactly as it appears in sudoers (if they are not that savvy, create them an alias).

Reply



Aaron Kili

August 19, 2019 at 2:16 pm

@Garry

These are practically better solutions, well explained. Am also testing them. Thanks for sharing.

Reply

Convert web pages and HTML files to PDF in your applications with the Pdfcrowd HTML to PDF API | Printed with Pdfcrowd.com